



University of Asia Pacific

Department of Computer Science and Engineering

CSE 438: Robotics Lab

Project Proposal

Project Title: Firefighting and Gas Detection Robot for Indoor Safety

Submitted by :

Team name: BotTrail

Member No	ID	Name
1	22201038	Tahiya Zareen
2	22201053	Rabea Sultana Shazia
3	22201052	Shah Abul Hasnat

Submitted to:

A S Zaforullah Momtaz

Assistant Professor,

Department of Computer Science and Engineering

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1. Introduction

Fire accidents and gas leakages are common hazards in homes, shops, and small businesses. These incidents can cause serious damage to property and may also threaten human life.

Most workplaces rely on basic safety systems such as fire extinguishers and smoke alarms, which require human presence. However, fires or gas leaks may occur when no one is present, leading to delayed response.

This project proposes a Firefighting and Gas Detection Robot that can automatically detect fire and gas leakage, provide alerts, and assist in extinguishing small fires.

2. Proposed Methodology

The development of the Firefighting and Gas Detection Robot will be carried out through several structured steps, combining hardware implementation and software control.

2.1 Hardware Setup

At the initial stage, all physical components of the robot will be assembled.

- A **PVC board or robot chassis** will be used as the base platform.
- Two **DC motors with wheels** will be mounted to enable movement.
- The **ESP32 microcontroller** will be installed as the main control unit.
- The **motor driver (L298N)** will be connected to control the motors.
- A **battery pack** will be used to supply power to all components.

All components will be securely mounted to ensure stability and proper functioning of the robot.

2.2 Sensor Integration

Sensors will be integrated to detect fire and gas leakage in the environment.

- The **MQ-2 gas sensor** will continuously monitor air quality and detect gases such as LPG, methane, and smoke.
- Multiple **flame sensors** will be placed at different positions (left, center, right) to detect the direction of fire.

- An optional **ultrasonic sensor** will be used to detect obstacles and measure distance.

Sensor data will be sent to the microcontroller for real-time processing.

2.3 Movement Control

The robot's movement system will be controlled using programmed logic.

- The **L298N motor driver** will receive signals from the microcontroller.
- Based on flame sensor input, the robot will **move toward the fire source**.
- The robot will be programmed to:
 - Move forward
 - Turn left or right depending on fire direction
 - Stop when close to the fire

This ensures accurate navigation toward the detected hazard.

2.4 Fire Extinguishing System

A fire suppression mechanism will be implemented to control small fires.

- A **12V water pump** will be connected using a **relay module**.
- When fire is detected, the microcontroller will activate the relay.
- The relay will turn on the pump, which will spray water through a pipe toward the fire.
- A **servo motor** may be used to adjust the direction of the water spray.

This allows the robot to actively extinguish small fires.

2.5 Alert System

An alert mechanism will be included to notify users about potential dangers.

- A **buzzer** will be activated when fire or gas leakage is detected.
- Optional **WiFi/mobile alerts** can be implemented using ESP32.
- Alerts will help users take immediate action even if they are not physically present.

2.6 System Integration

All subsystems will be combined into a single working unit.

- Sensors, motors, pump, and alert system will be connected to the microcontroller.
- The system will be programmed to perform tasks automatically.
- Proper wiring and power management will be ensured for smooth operation.

2.7 Testing and Calibration

The final stage involves testing and improving system performance.

- Individual components (sensors, motors, pump) will be tested separately.
- The complete system will be tested in real-life scenarios.
- Sensor sensitivity will be adjusted for accurate detection.
- Movement and response time will be optimized.

Any errors or issues will be fixed to ensure reliable performance

3. Social Impact and Motivation

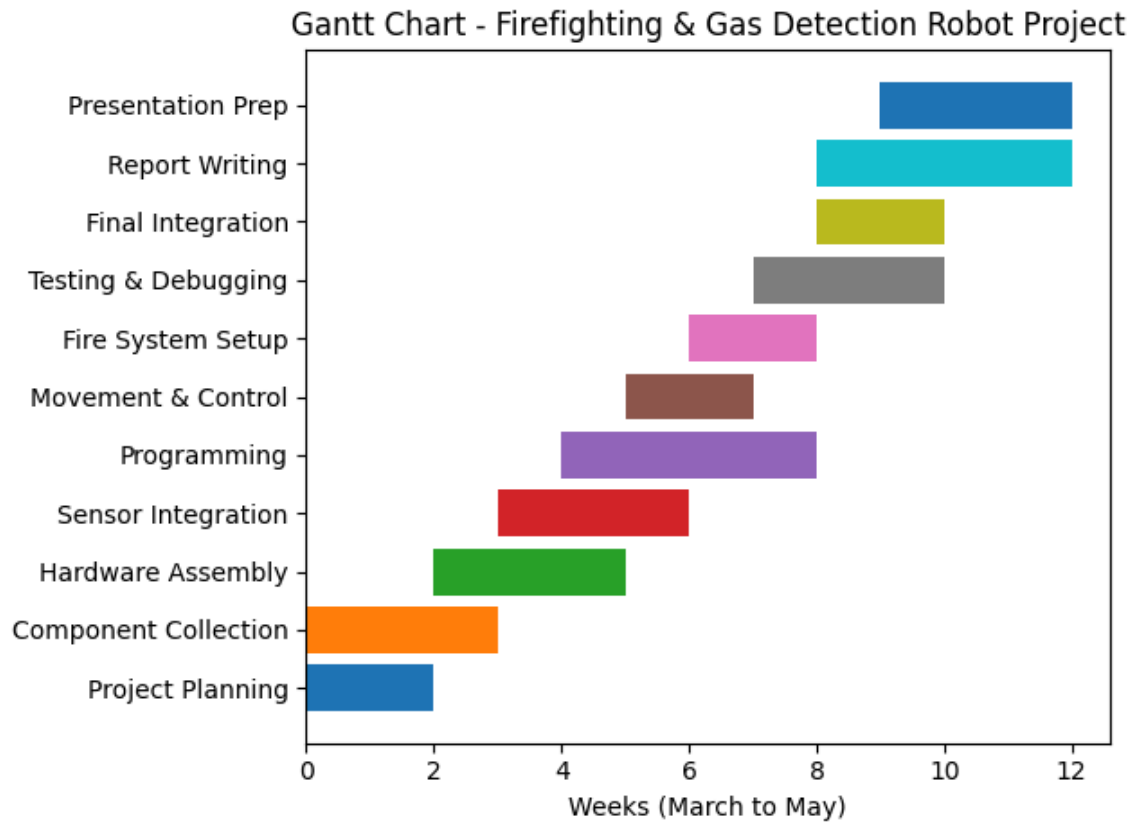
This project helps prevent fire accidents, protect lives and property, and provides a low-cost safety solution for small businesses. It enables automatic monitoring without human presence.

4. Budget Plan

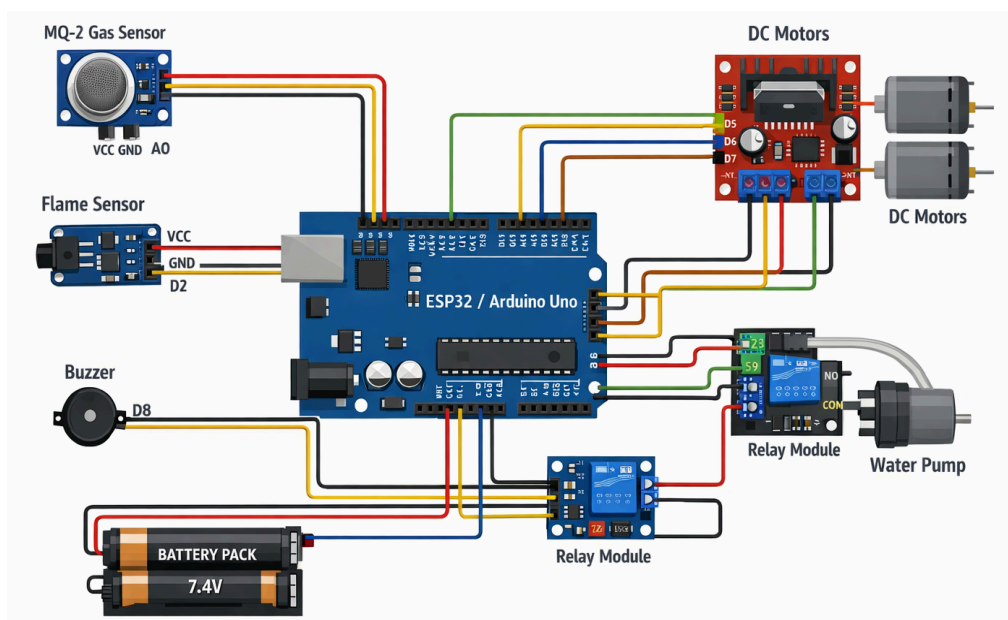
No.	Component	Quantity	Approx. Price (BDT)	Purpose / Notes
1	ESP32 Microcontroller	1	~500–700	Main controller to process sensor data and control motors, pump, and alerts
2	L298N Motor Driver Module	1	160	Controls direction and speed of DC motors

3	12V DC Motors with Wheels	2	$160 \times 2 = 320$	Provides movement of the robot
4	Servo Motor (SG90)	1	140	Rotates nozzle or sensor for better coverage
5	5V Relay Module (Single Channel)	1	80	Controls water pump safely using microcontroller
6	12V Mini Water Pump	1	130	Sprays water to extinguish fire
7	Flame Sensor Modules	3	$70 \times 3 = 210$	Detects fire direction (Left, Center, Right)
8	MQ-2 Gas Sensor	1	200	Detects gas leakage and smoke
9	Ultrasonic Sensor (HC-SR04)	1	100	Detects obstacles and measures distance
10	PVC Board / Chassis	1	240	Base platform for mounting all components
11	12V Battery Pack (3×18650 Li-ion)	1	80	Main power source
12	Battery Holder (18650)	2	170	Holds and connects battery cells safely
13	On/Off Switch	1	15	Controls power supply
14	Water Pipe (Silicone Tube)	1	20	Connects pump to nozzle

5. Gantt Chart



6. Circuit Diagram



7. CEP Mapping

Ks (Knowledge)

K	Description	How Addressed
K3	Engineering fundamentals	Use of sensors, circuits, and programming
K4	Specialized knowledge	Robotics system design and automation
K5	Design knowledge	Designing robot structure and control logic
K6	Engineering practice	Implementation using real hardware

Ps (Problem Solving)

P	Description	How Addressed
P1	Depth of knowledge	Integration of sensors + robotics
P2	Conflicting requirements	Cost vs performance balance
P7	Interdependence	Sensors, motors, and control system working together

As (Engineering Activities)

A	Description	How Addressed
A1	Resources	Use of hardware components and tools

A2	Interaction	Team collaboration and system integration
A5	Familiarity	Applying known robotics concepts in a new system

8. Conclusion

The proposed Firefighting and Gas Detection Robot aims to provide a **low-cost, automated safety solution** for detecting fire and gas leakage.

The system will help reduce risks, improve safety, and demonstrate the practical application of robotics in real-life problems. With proper implementation, this project can be extended into a more advanced smart safety system.